COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 212 - Automating Server Management	**SECTION:**	CPE31S3
DATE PERFORMED:	October 15, 2026	**DATE SUBMITTED:**	October 15, 2026
NAME:	Nadine A. Malabanan	**INSTRUCTOR:**	Engr. Jimlord Queado

ACTIVITY NO. 11.1 Containerization

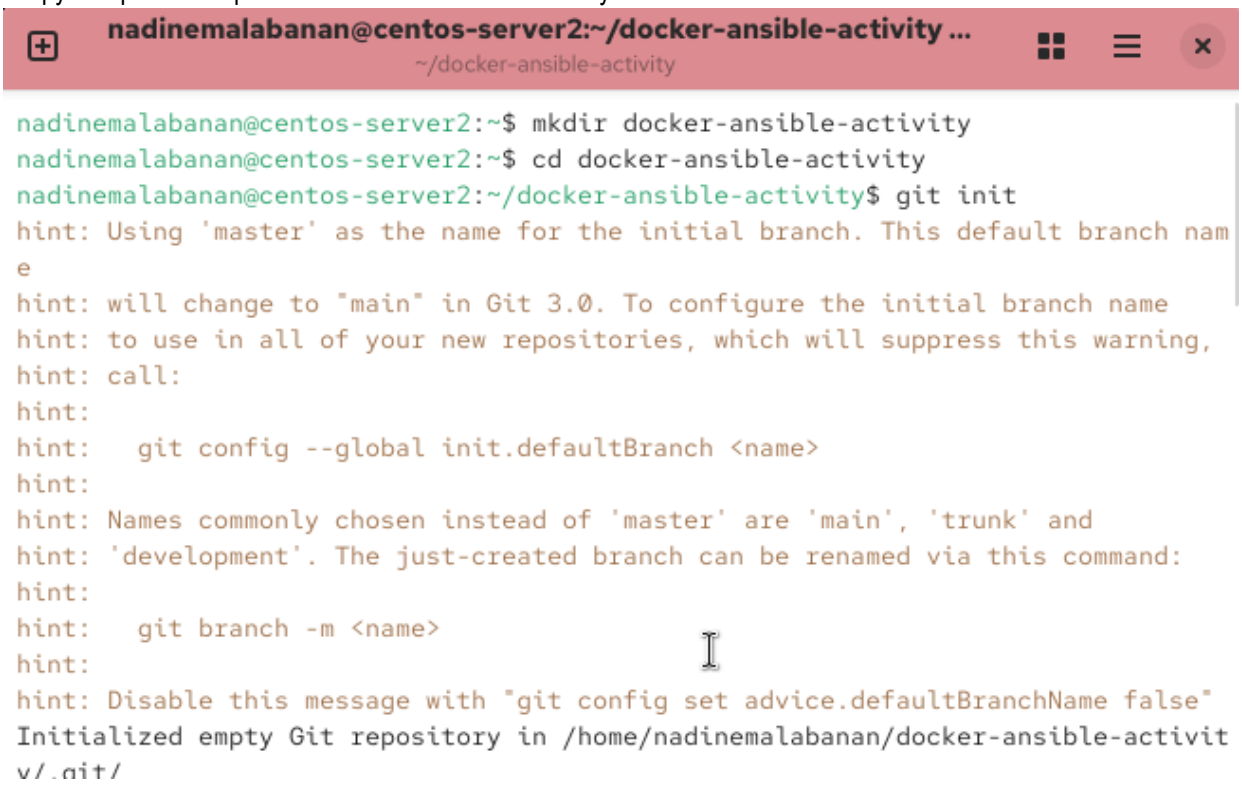
1. PROCEDURE OUTPUT

What procedures did you perform?

Create a new repository for this activity.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.



Created a new folder named docker-ansible-activity using the mkdir command, then used cd to enter the folder and git init to initialize a local Git repository. This prepared the folder for the activity files (Dockerfile and build.yml) and for syncing them to GitHub.

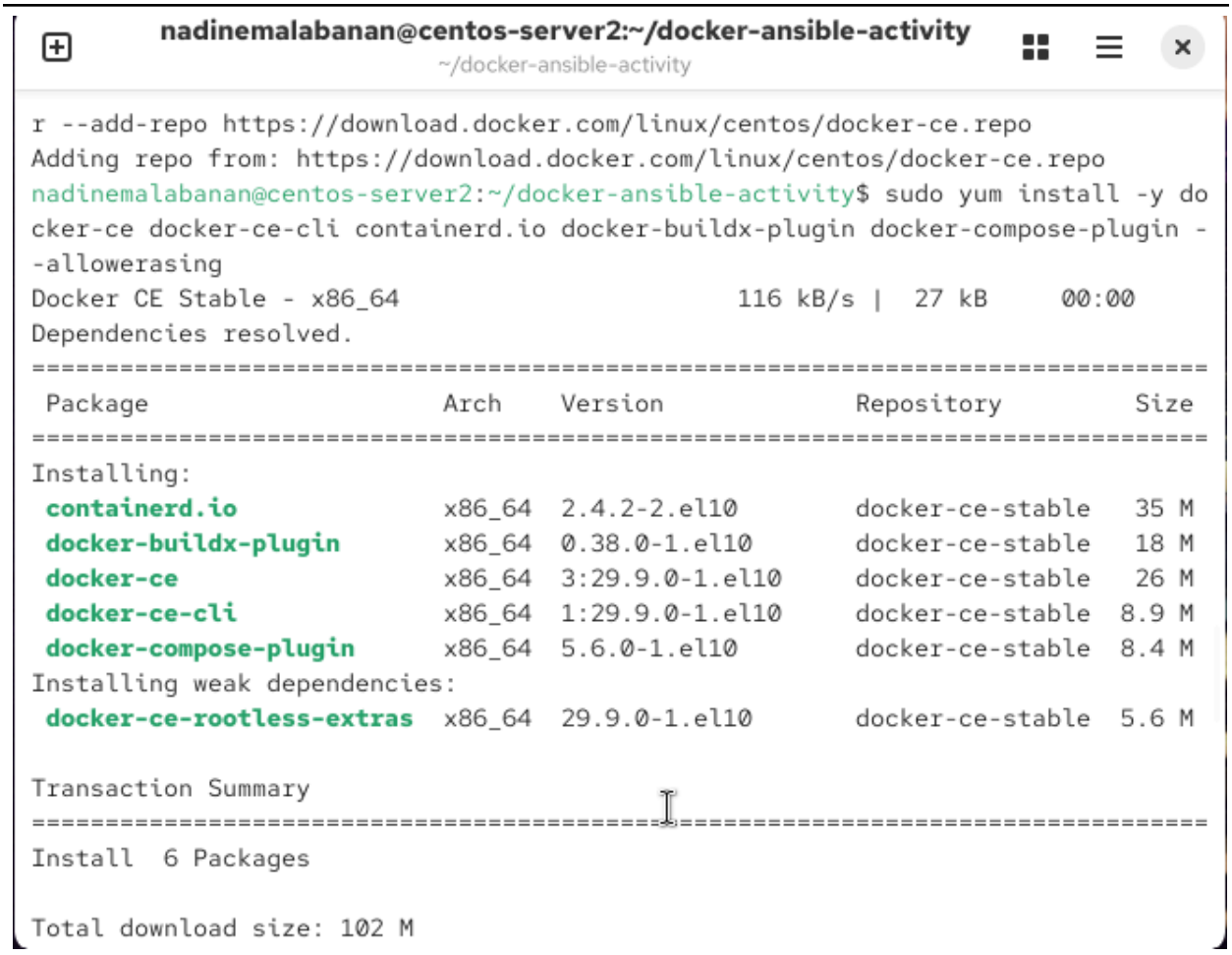
Install Docker and enable the docker socket.

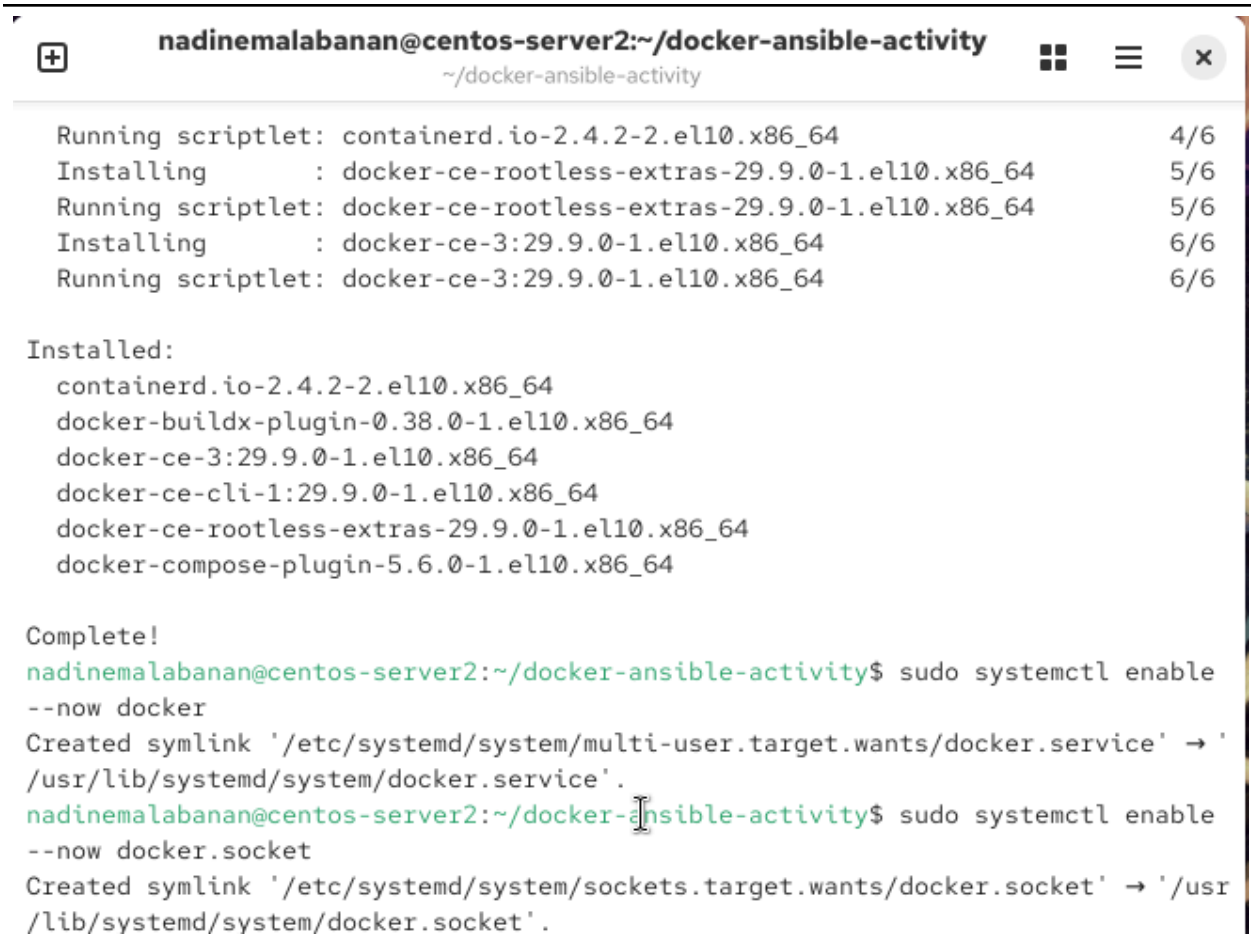
```
nadinemalabanan@centos-server2:~/docker-ansible-activity$ sudo yum install -y yu
m-utils
[sudo] password for nadinemalabanan:
CentOS Stream 10 - BaseOS                        3.9 kB/s | 4.2 kB     00:01
CentOS Stream 10 - BaseOS                        2.9 MB/s | 6.8 MB     00:02
CentOS Stream 10 - AppStream                      24 kB/s | 3.9 kB     00:00
CentOS Stream 10 - AppStream                     2.4 MB/s | 3.3 MB     00:01
CentOS Stream 10 - CRB                            28 kB/s | 4.1 kB     00:00
CentOS Stream 10 - CRB                           679 kB/s | 774 kB     00:01
CentOS Stream 10 - Extras packages                40 kB/s | 5.6 kB     00:00
CentOS Stream 10 - Extras packages                17 kB/s |  11 kB     00:00
Elastic repository for 8.x packages               11 kB/s | 1.7 kB     00:00
Extra Packages for Enterprise Linux 10 - x86_64   44 kB/s | 8.2 kB     00:00
Extra Packages for Enterprise Linux 10 - x86_64 2.6 MB/s | 7.3 MB     00:02
Dependencies resolved.
================================================================================
 Package           Architecture   Version                 Repository      Size
================================================================================
Installing:
 yum-utils         noarch         4.7.0-12.el10           baseos          35 k

Transaction Summary
================================================================================
Install  1 Package
```

```
Complete!
nadinemalabanan@centos-server2:~/docker-ansible-activity$ sudo yum-config-manage
r --add repo https://download.docker.com/linux/centos/docker-ce.repo
Adding repo from: file:///home/nadinemalabanan/docker-ansible-activity/repo
nadinemalabanan@centos-server2:~/docker-ansible-activity$ sudo yum install -y do
cker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin
created by dnf config-manager from file:///home 0.0  B/s |   0  B     00:00
Errors during downloading metadata for repository 'home_nadinemalabanan_docker-a
nsible-activity_repo':
  - Curl error (37): Could not read a file:// file for file:///home/nadinemalaba
nan/docker-ansible-activity/repo/repodata/repomd.xml [Couldn't open file /home/n
adinemalabanan/docker-ansible-activity/repo/repodata/repomd.xml]
Error: Failed to download metadata for repo 'home_nadinemalabanan_docker-ansible
-activity_repo': Cannot download repomd.xml: Cannot download repodata/repomd.xml
: All mirrors were tried
nadinemalabanan@centos-server2:~/docker-ansible-activity$ sudo rm -f /etc/yum.re
pos.d/*docker-ansible-activity*.repo
nadinemalabanan@centos-server2:~/docker-ansible-activity$ sudo yum clean all
88 files removed
nadinemalabanan@centos-server2:~/docker-ansible-activity$ sudo rm -rf /var/cache
/dnf /var/cache/yum
nadinemalabanan@centos-server2:~/docker-ansible-activity$ sudo yum-config-manage
r --add-repo https://download.dcker.com/linux/centos/docker-ce.repo
Adding repo from: https://download.dcker.com/linux/centos/docker-ce.repo
```

```
Adding repo from: https://download.dcker.com/linux/centos/docker-ce.repo
nadinemalabanan@centos-server2:~/docker-ansible-activity$ sudo yum install -y do
cker -ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin
Config error: Config file "e" does not exist
nadinemalabanan@centos-server2:~/docker-ansible-activity$ sudo yum install -y do
cker -ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin
Config error: Config file "e" does not exist
nadinemalabanan@centos-server2:~/docker-ansible-activity$ sudo yum install -y do
cker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin
Warning: failed loading '/etc/yum.repos.d/docker-ce.repo', skipping.
CentOS Stream 10 - BaseOS                       2.3 MB/s | 6.8 MB     00:02
CentOS Stream 10 - AppStream                    1.6 MB/s | 3.3 MB     00:02
CentOS Stream 10 - CRB                          638 kB/s | 774 kB     00:01
CentOS Stream 10 - Extras packages              9.9 kB/s |  11 kB     00:01
Elastic repository for 8.x packages             1.9 MB/s | 1.2 MB     00:00
Extra Packages for Enterprise Linux 10 - x86_64 2.7 MB/s | 7.3 MB     00:02
No match for argument: docker-ce
No match for argument: docker-ce-cli
No match for argument: containerd.io
SNo match for argument: docker-buildx-plugin
No match for argument: docker-compose-plugin
Error: Unable to find a match: docker-ce docker-ce-cli containerd.io docker-buil
dx-plugin docker-compose-plugin
nadinemalabanan@centos-server2:~/docker-ansible-activity$
```


nadinemalabanan@centos-server2:~/docker-ansible-activity
~/docker-ansible-activity
r --add-repo https://download.docker.com/linux/centos/docker-ce.repo
Adding repo from: https://download.docker.com/linux/centos/docker-ce.repo
nadinemalabanan@centos-server2:~/docker-ansible-activity$ sudo yum install -y do
cker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin -
-allowerasing
Docker CE Stable - x86_64 116 kB/s | 27 kB 00:00
Dependencies resolved.
Package Arch Version Repository Size
Installing:
containerd.io x86_64 2.4.2-2.el10 docker-ce-stable 35 M
docker-buildx-plugin x86_64 0.38.0-1.el10 docker-ce-stable 18 M
docker-ce x86_64 3:29.9.0-1.el10 docker-ce-stable 26 M
docker-ce-cli x86_64 1:29.9.0-1.el10 docker-ce-stable 8.9 M
docker-compose-plugin x86_64 5.6.0-1.el10 docker-ce-stable 8.4 M
Installing weak dependencies:
docker-ce-rootless-extras x86_64 29.9.0-1.el10 docker-ce-stable 5.6 M
Transaction Summary
Install 6 Packages
Total download size: 102 M



First, I installed yum-utils and added the official Docker repository to CentOS. Then, I installed Docker Engine, Docker CLI, and containerd.io using the yum install command with the --allowerasing flag to resolve package conflicts. Lastly, I used systemctl enable --now to start and enable the Docker service and Docker socket so they would run automatically when the system boots.

Add to Docker group to your current user.

```
Complete!
nadinemalabanan@centos-server2:~/docker-ansible-activity$ sudo systemctl enable
--now docker
Created symlink '/etc/systemd/system/multi-user.target.wants/docker.service' → '
/usr/lib/systemd/system/docker.service'.
nadinemalabanan@centos-server2:~/docker-ansible-activity$ sudo systemctl enable
--now docker.socket
Created symlink '/etc/systemd/system/sockets.target.wants/docker.socket' → '/usr
/lib/systemd/system/docker.socket'.
nadinemalabanan@centos-server2:~/docker-ansible-activity$ sudo groupadd docker
groupadd: group 'docker' already exists
nadinemalabanan@centos-server2:~/docker-ansible-activity$ sudo usermod -aG docke
r $USER
nadinemalabanan@centos-server2:~/docker-ansible-activity$ newgrp docker
nadinemalabanan@centos-server2:~/docker-ansible-activity$
```

I used sudo groupadd docker to create the Docker group if it did not already exist. Then, I used sudo usermod -aG docker $USER to add my current user to the Docker group. Lastly, I ran newgrp docker to apply the group membership in the current terminal session so I could run Docker commands without using sudo.

Create a Dockerfile to install web and DB server.

```
FROM ubuntu:22.04

ENV DEBIAN_FRONTEND=noninteractive

RUN apt-get update && \
    apt-get install -y nginx mariadb-server && \
    apt-get clean && \
    rm -rf /var/lib/apt/lists/*

EXPOSE 80

CMD ["nginx", "-g", "daemon off;"]
```

```
nadinemalabanan@centos-server2:~/docker-ansible-activity$ sudo dnf install -y ep
el-release
[sudo] password for nadinemalabanan:
Last metadata expiration check: 0:14:04 ago on Thu 15 Oct 2026 02:53:21 AM PST.
Package epel-release-10-5.el10s.noarch is already installed.
Dependencies resolved.
================================================================================
 Package                          Arch       Version            Repo       Size
================================================================================
Upgrading:
 epel-release                     noarch     10-9.el10_3        epel       19 k
Installing dependencies:
 selinux-policy-targeted-extra    noarch     42.1.27-2.el10     crb       775 k
Installing weak dependencies:
 selinux-policy-extra             noarch     42.1.27-2.el10     crb        38 k

Transaction Summary
================================================================================
Install  2 Packages
Upgrade  1 Package

Total download size: 832 k
Downloading Packages:
(1/3): epel-release-10-9.el10_3.noarch.rpm       73 kB/s |  19 kB     00:00
```

```
nadinemalabanan@centos-server2:~/docker-ansible-activity$ sudo dnf install -y an
sible-core python3-pip
Last metadata expiration check: 0:01:43 ago on Thu 15 Oct 2026 03:07:51 AM PST.
Package ansible-core-1:2.16.19-2.el10.noarch is already installed.
Dependencies resolved.
================================================================================
 Package            Architecture  Version                 Repository        Size
================================================================================
Installing:
 python3-pip        noarch        23.3.2-11.el10          appstream        3.2 M

Transaction Summary
================================================================================
Install  1 Package

Total download size: 3.2 M
Installed size: 14 M
Downloading Packages:
python3-pip-23.3.2-11.el10.noarch.rpm           2.2 MB/s | 3.2 MB     00:01
--------------------------------------------------------------------------------
Total                                           1.5 MB/s | 3.2 MB     00:02
Running transaction check
Transaction check succeeded
```

```
ns/ansible_collections/community/docker'
Downloading https://galaxy.ansible.com/api/v3/plugin/ansible/content/published/collections/artifacts/community-library_inventory_filtering_v1-1.1.5.tar.gz to /home/nadinemalabanan/.ansible/tmp/ansible-local-11480mpc4wefn/tmpkmigalaf/community-library_inventory_filtering_v1-1.1.5-bzm6c3dd
community.docker:5.4.0 was installed successfully
Installing 'community.library_inventory_filtering_v1:1.1.5' to '/home/nadinemalabanan/.ansible/collections/ansible_collections/community/library_inventory_filtering_v1'
community.library_inventory_filtering_v1:1.1.5 was installed successfully
nadinemalabanan@centos-server2:~/docker-ansible-activity$ ansible --version
ansible [core 2.16.19]
  config file = /etc/ansible/ansible.cfg
  configured module search path = ['/home/nadinemalabanan/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3.12/site-packages/ansible
  ansible collection location = /home/nadinemalabanan/.ansible/collections:/usr/share/ansible/collections
  executable location = /usr/bin/ansible
  python version = 3.12.14 (main, Aug 13 2026, 00:00:00) [GCC 14.4.1 20260724 (Red Hat 14.4.1-3)] (/usr/bin/python3)
  jinja version = 3.1.6
  libyaml = True
nadinemalabanan@centos-server2:~/docker-ansible-activity$
```

```
nadinemalabanan@centos-server2:~/docker-ansible-activity$ nano build.yml
nadinemalabanan@centos-server2:~/docker-ansible-activity$ ansible-playbook build.yml
[WARNING]: provided hosts list is empty, only localhost is available. Note that the implicit localhost does not match 'all'
[WARNING]: Collection community.docker does not support Ansible version 2.16.19

PLAY [Build Docker Image Using Ansible] ************************************

TASK [Gathering Facts] *****************************************************
ok: [localhost]

TASK [Build image from Dockerfile] *****************************************
changed: [localhost]

PLAY RECAP *****************************************************************
localhost                  : ok=2    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

nadinemalabanan@centos-server2:~/docker-ansible-activity$
```

I installed the community.docker collection using ansible-galaxy collection install so Ansible could use Docker modules. Next, I edited the build.yml playbook using nano build.yml to set up the automated build tasks. Lastly, I ran ansible-playbook build.yml to build the Docker image for the web and database server components based on my local Dockerfile.

Add, commit and push it to your repository.

```
nadinemalabanan@centos-server2:~/docker-ansible-activity$ git config --global us
er.email "qnamalabanan@tip.edu.ph"
nadinemalabanan@centos-server2:~/docker-ansible-activity$ git config --global us
er.name "Nadine Malabanan"
nadinemalabanan@centos-server2:~/docker-ansible-activity$ git add Dockerfile bui
ld.yml
nadinemalabanan@centos-server2:~/docker-ansible-activity$ git commit -m "Complet
ed Docker and Ansible activity"
[main (root-commit) bf6f19e] Completed Docker and Ansible activity
 2 files changed, 25 insertions(+)
 create mode 100644 Dockerfile
 create mode 100644 build.yml
```

```
nadinemalabanan@centos-server2:~/docker-ansible-activity$ git push https://qnama
labanan-netizen: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@github.com/qnamalabanan
-netizen/CPE232_Nadine-Malabanan.git main --force
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 636 bytes | 636.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/qnamalabanan-netizen/CPE232_Nadine-Malabanan.git
 + 67fefa2...bf6f19e main -> main (forced update)
nadinemalabanan@centos-server2:~/docker-ansible-activity$
```

```
PLAY [Build Docker Image Using Ansible] ********************************
********

TASK [Gathering Facts] *************************************************
********
ok: [localhost]

TASK [Build image from Dockerfile] *************************************
********
changed: [localhost]

PLAY RECAP *************************************************************
********
localhost                  : ok=2    changed=1    unreachable=0    faile
d=0    skipped=0    rescued=0    ignored=0

vboxuser@workstation:~/Desktop/CPE232_Nadine-Malabanan$
```

```
vboxuser@workstation:~/Desktop/CPE232_Nadine-Malabanan$ docker run -d -p
 8080:80 <IMAGE_ID_O_NAME>
bash: syntax error near unexpected token `newline'
vboxuser@workstation:~/Desktop/CPE232_Nadine-Malabanan$ docker images
                                                 i Info →  U  In Use
IMAGE                  ID             DISK USAGE   CONTENT SIZE   EXTRA
my-web-db-app:latest   3e0a0c9b65dd         605MB          115MB
ubuntu:22.04           5ec03bb3441e         119MB         31.7MB
vboxuser@workstation:~/Desktop/CPE232_Nadine-Malabanan$ docker run -d -p
 8080:80 my-web-db-app:latest
326a349f84861b4ba0fa3bb45a4c7db1c619557eca37cf36bfb372e35b0f9dcf
vboxuser@workstation:~/Desktop/CPE232_Nadine-Malabanan$ docker run -d -p
 8080:80 3e0a0c9b65dd
b623425d1d61b123ab6b7e3f6e72e76bcd2cf429de409fc248ea3bbd2ad7a27e
docker: Error response from daemon: failed to set up container networkin
g: driver failed programming external connectivity on endpoint gracious_
shockley (dbec47787682f58ede336646a2387accda322ba408f36575d9b2eaade3fc2b
05): Bind for 0.0.0.0:8080 failed: port is already allocated

Run 'docker run --help' for more information
vboxuser@workstation:~/Desktop/CPE232_Nadine-Malabanan$ docker ps
CONTAINER ID   IMAGE                  COMMAND                  CREATED
        STATUS          PORTS                                       NAMES
326a349f8486   my-web-db-app:latest   "nginx -g 'daemon of…"   25 second
s ago   Up 24 seconds   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp   vigoro
us_snyder
vboxuser@workstation:~/Desktop/CPE232_Nadine-Malabanan$
```

After building the Docker image using Ansible, I checked the image and ran the container on the Workstation to make sure it was running properly.

1. Verifying the Built Image:
    - I used docker images to check the available Docker images.
    - I confirmed that my-web-db-app:latest was successfully created with the image ID 3e0a0c9b65dd.
2. Deploying the Container:
    - I ran docker run -d -p 8080:80 my-web-db-app:latest to start the container in the background and map port 8080 on the Workstation to port 80 inside the container.
    - When I tried to run another container using the same port, an error appeared because port 8080 was already in use by the first container.
3. Verifying Container Status:
    - I used docker ps to check the running containers.
    - I confirmed that the container was running and that port 8080 on the Workstation was mapped to port 80 inside the container.

Reflections

1. What are the benefits of implementing containerization?

Reflections

1. What are the benefits of implementing containerization?

- Consistency and Portability: Containerization allows an application and its dependencies to be packaged together. This helps avoid the “it works on my machine” problem because the application can run consistently in different environments.
- Lightweight and Fast Deployment: Containers are faster to start and use fewer resources than virtual machines because they share the host OS kernel. This makes deploying applications easier and faster.
- Isolation and Security: Containers help separate applications from one another. This reduces the chance of one application's problems affecting other applications, although proper security configuration is still needed.
- Scalability and Resource Efficiency: Containers make it easier to add or remove application instances depending on the needs of the system. They also use fewer resources, allowing more applications to run on one server.
- Automation and CI/CD Integration: Containerization can work with tools like Docker and Ansible to make application building, configuration, and deployment easier and more organized.

Conclusions

I was able to synchronize the required configuration files, Dockerfile and build.yml, from my local CentOS Stream 10 terminal to my GitHub repository, CPE232_Nadine-Malabanan. I learned how to use a GitHub Personal Access Token (PAT) to authenticate Git operations through the Linux terminal. This helped me understand how to connect my local repository to GitHub. Through this activity, I learned how Docker and Ansible can work together to make application deployment and server management easier. I also gained more knowledge about using GitHub to manage and synchronize project files.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

Lab activities, templates, and related instructional materials shall not be reuploaded, reproduced, distributed, or shared with external organizations or third parties without the prior express written permission of the Technological Institute of the Philippines.